

# ZHIXIANG FAN

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Design PhD student working on spatial intelligence and interactive world modeling, with a background in landscape architecture. Builds systems for semantic 3D reconstruction, agent-driven scene creation and immersive interaction, translating spatial design knowledge into computational representations and tools.

## EDUCATION

### Tsinghua University · Academy of Arts & Design

Sep 2024–Present

PhD student in Design. Research: 3D reconstruction, spatial intelligence and human–AI interaction.

### Nanjing Forestry University

Sep 2021–Jun 2024

Master's in Landscape Architecture. Research: spatial digitization and environmental analysis.

## RESEARCH AND INDUSTRY EXPERIENCE

### Beijing Qijie Interactive · World Model Technology & Product Lead

Mar 2026–Present

Lead technical development of Axis, a 3D world creation platform. Built server deployments, model integrations, the web 3D runtime and generation workflows from scratch. [Product demo](#)

Coordinate generation APIs and agent task states for Minecraft scenes; extending scene representations and interaction to other environments and WebXR.

### Beijing Yuaiweiwu · AI Product & Algorithm Intern

Jun–Aug 2026

Evaluated tutoring dialogues for answer guessing, spoken formulas and long-context behavior. Built resumable extraction and validation for 592 conversations and 9,072 messages from ~11 GB of JSONL data; translated failures into model and product feedback.

### Dongfeng R&D Institute · AI Center Intern

Jun–Sep 2025

Built Wuhan road networks and 3D simulation scenes with OSM, RoadRunner and Blender for UE4/CARLA. Prototyped a perception–decision–control loop and designed scenario tests for end-to-end driving models.

## SELECTED RESEARCH PROJECTS

### Semantic 3D Gaussian Splatting and Language Interaction

Feb 2026

Integrated SAM, CLIP, LangSplat and ResGS for open-vocabulary 3D semantics. Extracted object locations with DBSCAN and built a Three.js interface for real-time rendering of million-scale Gaussians, semantic picking and LLM spatial interaction.

### Immersive World Creation in VR

Apr 2026

Connected text/image-to-world generation with browser-based WebXR and 6-DoF navigation. Deployed the generation-to-exploration workflow in projects including a Tsinghua psychology summer program.

### Design Skills for Agent-Driven Scene Generation

Jun 2026

Encoded Chinese garden design principles as composable rules for regions, paths, sightlines, scale and spatial sequences.

Implemented WorldSpec, skill compilation, constraint checking and local repair, with parameter editing, region locking and inspectable design rationales.

### Hybrid 3DGS–Mesh–Physics Scene Editing

Jul 2026–Present

Combine reconstructed Gaussian scenes with mesh proxies, colliders and physical properties. Investigating speech, pointing and direct manipulation for reversible scene edits, with checks on scale, support, collision and reachability.

## SELECTED PUBLICATIONS

### A Zero-Parameter EEG-fNIRS Rule Ensemble for Cross-Subject Emotion Tracking.

ACM Multimedia 2026 · MER Challenge (*accepted*). Team lead; organized experiments, analyzed results and wrote the paper. Rule ensemble achieved overall MAE 25.52 (valence 24.62; arousal 26.41).

### Endless Life of Stone: An Embodied Journey into the Universe within the Void.

ACM Multimedia 2026 · Interactive Art (*accepted*). Interactive installation combining AI-generated video with camera/MediaPipe audience-distance tracking to control the video timeline.

Y. Huang, Z. Fan, et al. [Research on the spatial distribution and influencing factors of hydronyms in cities along the Grand Canal.](#) *Humanities and Social Sciences Communications*, 2026.

LLM-assisted semantic classification of 10,000+ spatial records, with GIS and spatial statistical analysis.

Y. Huang, Z. Fan, et al. [Evaluating visual experience in heritage-rich areas: A case study of the Inner Ring Road of West Lake, Hangzhou.](#) *Frontiers of Architectural Research*, 2026.

Applied DeepLab v3 to street-view images; quantified greenery and enclosure and analyzed spatial patterns.

Z. Fan, T. Song and X. Tang. [The Spatial Relationship between Cities and Mountains in Jiangnan Region in the Qing Dynasty Based on Historical Texts.](#) *ACM Journal on Computing and Cultural Heritage*, 2024.

Extracted geographic entities and spatial relations from historical texts for GIS reconstruction and visualization.

## TECHNICAL SKILLS

**Programming and systems:** Python, JavaScript, PyTorch, basic CUDA, Linux/cloud deployment, API integration and agent workflows.

**3D and interaction:** 3DGS, SAM/CLIP, Three.js/WebGL, Blender, UE/CARLA, WebXR, spatial design and procedural scene creation.